



Drive Test EDA Cleaning Outputs Report

LTE/WCDMA QoE, Events, Anomaly Detection, and RCA Direction

Analysis of Cleaned Output Tables and Recommended Data
Improvements

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Contents

1	Executive Summary	2
1.1	Files Covered	2
2	Recommended Project Direction	4
3	Raw Schema and Cleaning Control Outputs	4
3.1	drive_test_data_dictionary.csv	4
3.2	drive_test_datetime_candidates.csv	5
3.3	detected_key_columns.csv	6
3.4	cleaned_data_dictionary.csv	6
3.5	removed_columns_report.csv	7
4	Activity and Event Outputs	8
4.1	activity_table.csv	8
4.2	events_long.csv	10
4.3	event_summary_Events_titles.csv	11
5	Planning Validation Output	12
5.1	drive_nearest_planning_sample.csv	12
6	Technology Performance Summaries	13
6.1	lte_summary_by_pci.csv	13
6.2	wcdma_summary_by_psc.csv	14
7	Session-Level Output	15
7.1	session_summary.csv	15
8	Cleaned Row-Level Dataset	17
8.1	drive_clean_base.parquet	17
9	How These Outputs Improve the Cleaned Data	18
10	Recommended Removed Columns to Bring Back Later	18
11	Final Recommendation	20

1 Executive Summary

This report documents the cleaned EDA outputs generated from the drive-test dataset. The cleaning strategy used a 70% missing-value threshold while protecting important time, location, activity, and event fields. The current evidence shows that the usable measurements are mainly LTE, WCDMA, activity, event, and GPS fields. Therefore, the strongest project direction is **LTE/WCDMA Drive-Test QoE Anomaly Detection and Root Cause Analysis**.

Main conclusion

The report outputs support LTE throughput modeling, LTE carrier aggregation optimization, event-window RCA, MO call/CSFB analysis, WCDMA quality analysis, and route-based anomaly mapping. Direct 5G NR PCI/RSI planning validation should not be the main project direction unless a matching Fiji/Nadi NR planning file and usable NR measurements are obtained.

1.1 Files Covered

File	Rows	Cols	Size	Role
drive_test_data_dictionary.csv	1,365	15	204.3 KB	Raw schema and missingness profile for the original drive-test file.
drive_test_datetime_candidates.csv	25	6	1.6 KB	Detected timestamp-like columns and parse validation.
detected_key_columns.csv	17	2	0.6 KB	Auto-selected canonical telecom KPI columns.
cleaned_data_dictionary.csv	182	15	39.8 KB	Schema profile of the cleaned base dataset.
removed_columns_report.csv	1,183	17	182.2 KB	Audit trail of dropped columns and removal reasons.
activity_table.csv	465	16	138.5 KB	One row per call or HTTP download activity.
events_long.csv	102,302	12	36.5 MB	One row per unpacked event from the original packed event fields.
event_summary_Events_titles.csv	117	2	2.8 KB	Frequency summary of event titles.
drive_nearest_planning_sample.csv	20,000	22	4.8 MB	Geographic check between drive route and planning file.
lte_summary_by_pci.csv	190	9	26.5 KB	LTE serving PCI performance summary.
session_summary.csv	546	28	182.1 KB	One row per drive-test session/id.
wcdma_summary_by_psc.csv	168	6	12.0 KB	WCDMA serving PSC performance summary.
drive_clean_base.parquet	64,949	212	18.7 MB	Cleaned row-level dataset stored in Parquet for fast analytics and ML.

Table 1: Output files analyzed in this report. Table columns use wrapped text to avoid overflow.

2 Recommended Project Direction

The outputs suggest two clean project branches. The first branch is an LTE download-performance branch using LTE KPIs, throughput, CA status, events, and location. The second branch is a WCDMA/MO-call branch using WCDMA RSCP/EcNo, CSFB/call RAT, call activity status, setup timing, and event windows.

Recommended project title

Drive-Test QoE Anomaly Detection and Root Cause Analysis for LTE/WCDMA Mobility Networks

Recommended modules:

- **LTE throughput prediction:** predict expected downlink throughput from radio, scheduling, CA, and resource features.
- **Throughput anomaly detection:** flag samples, sessions, or PCIs where real throughput is much lower than expected.
- **Carrier aggregation optimization:** evaluate whether CA activation, carrier count, and SCell quality improve throughput.
- **Event-window RCA:** inspect KPI behavior before and after RACH, handover, CA, and WCDMA active-set events.
- **MO call and CSFB RCA:** explain call success/failure using WCDMA quality, call setup timing, events, and session-level context.
- **Route dashboard:** map RSRP, SINR, throughput, PCI/PSC, event density, and bad sessions.

3 Raw Schema and Cleaning Control Outputs

3.1 drive_test_data_dictionary.csv

This file is the schema profile of the original drive-test file before cleaning. It is important because it proves which fields are empty, sparse, constant, numeric, categorical, or useful for telecom analysis.

Parameter / Column	Meaning
column	Original raw column name.
dtype	Detected pandas data type.
full_non_null_count	Number of populated rows in the full dataset.
full_missing_pct	Missing percentage used by the 70% cleaning rule.
sample_unique_count	Approximate cardinality, useful for detecting constant/repeated fields.
sample_non_null_examples	Example values that help interpret unknown vendor columns.
sample_min / sample_max / sample_mean / sample_std	Numeric profiling fields for radio KPIs and counters.

Table 2: Important fields in drive_test_data_dictionary.csv.

Importance and use. This file controls the whole cleaning process. It helps identify which columns should be removed, protected, or extracted into separate tables. It also helps detect project opportunities such as LTE KPIs, WCDMA KPIs, event fields, activity fields, and location fields.

Cleaning improvement. Keep using this table as the audit source before every notebook rebuild. Add a column called `cleaning_action` later, with values such as keep, drop, protect, restore, or convert-to-long-table.

3.2 drive_test_datetime_candidates.csv

This file lists timestamp-like columns and validates whether they can be parsed into datetime values. It is essential because the project needs a reliable time axis for route progression, event alignment, and before/after analysis.

Parameter / Column	Meaning
column	Candidate timestamp column.
dtype	Original loaded type.
name_hint	Whether the column name suggests time/date information.
parse_success_pct	Percentage of sample values parsed successfully.
sample_min	Earliest detected timestamp.
sample_max	Latest detected timestamp.

Table 3: Important fields in drive_test_datetime_candidates.csv.

Importance and use. The main row-level clock is `timestamp`. Activity start/end timestamps

define activity windows, while event timestamps define network or script event timing. This enables event-window RCA, sample-gap checks, and session-duration validation.

Cleaning improvement. Keep parsed versions of the main timestamps and use `timestamp` for row-level sorting. Do not treat ID-derived session/log times as moving sample clocks; they are metadata for the whole session/log.

3.3 detected_key_columns.csv

This file records the notebook's detected key telecom columns. It is useful as a compact reference for canonical KPI selection.

Parameter / Column	Meaning
meaning	Human-readable KPI meaning such as LTE RSRP, LTE SINR, LTE DL throughput, WCDMA RSCP, or location.
column	Actual dataframe column selected for that meaning.

Table 4: Important fields in detected_key_columns.csv.

Importance and use. This file helps standardize analysis by avoiding repeated manual column selection. It also reveals whether important target variables survived cleaning.

Cleaning improvement. Later, use it to create renamed canonical columns such as `lte_rsrp`, `lte_sinr`, `lte_dl_throughput`, `wcdma_rscp`, and `wcdma_ecno`. This makes ML code simpler and less vendor-specific.

3.4 cleaned_data_dictionary.csv

This file describes the cleaned dataset after applying the 70% missingness threshold and protected-column logic. It shows which columns remain and what families they belong to.

Item	Value
LTE	114 columns
WCDMA	42 columns
Activity	10 columns
Events	6 columns
Location	5 columns
Extra_KPIs	2 columns
id	1 columns
imsi	1 columns
timestamp	1 columns

Table 5: Cleaned data family distribution.

Parameter / Column	Meaning
column	Column name in the cleaned base dataset.
family	Column group such as LTE, WCDMA, Activity, Events, Location, Extra_KPIs, or metadata.
full_missing_pct	Remaining missing percentage after cleaning.
sample_top_values	Top repeated values for categorical columns.
sample_min / sample_max / sample_mean / sample_std	Numeric profiling fields for KPI columns.

Table 6: Important fields in cleaned_data_dictionary.csv.

Importance and use. The cleaned dictionary shows that the useful data is mainly LTE, WCDMA, Activity, Events, Location, and Extra KPI fields. It also confirms that usable NR5G sample-level measurements did not survive/exist in this dataset.

Cleaning improvement. Use this table to create specialized output tables: LTE samples, WCDMA samples, event timeline, activity table, session summary, and route/location table. Avoid using one huge mixed table directly for all ML tasks.

3.5 removed_columns_report.csv

This file is the audit trail of columns removed by the cleaning rules. It is important because sparse telecom fields are not always useless; some are sparse because they occur only during events, handovers, calls, or HTTP tests.

Item	Value
>70% missing	634 columns
100% missing	518 columns
constant / single unique value	31 columns

Table 7: Removed-column reasons.

Parameter / Column	Meaning
column	Removed column name.
family	Column family inferred from the name.
full_non_null_count	Number of rows still containing useful values.
full_missing_pct	Reason the column may have been dropped by threshold.
is_protected	Whether the column was protected from removal.
remove_reason	Decision reason such as 100% missing, more than 70% missing, or constant/single unique value.

Table 8: Important fields in removed_columns_report.csv.

Importance and use. This file lets us recover meaningful sparse columns later. The most important candidates are HTTP throughput, handover interruption time, call setup timing, CSFB timing, and activity application details.

Cleaning improvement. Do not blindly restore all removed columns. Instead, restore selected sparse KPIs into an enriched table and convert neighbor/active-set groups into long tables.

4 Activity and Event Outputs

4.1 activity_table.csv

The activity table is an activity-level table, not a row-level sample table. Each row represents a detected call or HTTP/HTTPS download activity.

Item	Value
MO Call	274 activities
HTTP/HTTPS Download	191 activities
Status: Success	253 activities
Status: Failed	212 activities

Table 9: Activity distribution.

Parameter / Column	Meaning
id	Session/log identifier linking the activity to row-level samples.
timestamp	Sample timestamp associated with the activity record.
id_route_test_label	Parsed route and test label, e.g., Fiji-Nadi-MO-D.
id_test_type	Parsed test type, such as MO-D or DL-D.
Activity.Activity	Activity type, such as MO Call or HTTP/HTTPS Download.
Activity.Duration	Activity duration, originally in milliseconds.
Activity.Status	Success or Failed activity result.
Activity.Tech	Technology observed for the activity.
Activity.Power / Activity.Quality / Activity.SINR	Activity-level radio context.
Activity.Throughput	Activity-level throughput or call-related numeric result.
Activity.Start TimeStamp_parsed / Activity.End TimeStamp_parsed	Parsed activity window boundaries.

Table 10: Important fields in activity_table.csv.

Importance and use. This table is excellent for activity-level QoE labels. MO Call rows can support call success/failure classification. HTTP download rows can support throughput regression or application-layer anomaly detection.

Cleaning improvement. Keep activity columns protected, but model activities separately from row-level samples. Convert duration to seconds and avoid mixing MO call and HTTP download targets in one model.

4.2 events_long.csv

This file is one of the most important outputs. The raw event fields originally stored multiple events inside one cell using packed strings. The long event table converts that structure into a clean format where one row equals one event.

Item	Value
Network	93,873 events
Script	5,930 events
Logs	2,468 events
Device	31 events

Table 11: Event category distribution.

Item	Value
LTE	76,771 events
WCDMA	24,895 events
GSM	636 events

Table 12: Event technology distribution.

Parameter / Column	Meaning
<code>source_index</code>	Original row index in the cleaned row-level dataset where the event was found.
<code>event_order_in_sample</code>	Order of the event inside the original packed sample row.
<code>id</code>	Session/log identifier used for joining events with sessions and samples.
<code>sample_timestamp</code>	Timestamp of the sample row that carried the event.
<code>Events.Timestamp</code>	Original event timestamp before parsing.
<code>Events.category</code>	High-level event category such as Network, Script, Logs, or Device.
<code>Events.Tech</code>	Technology associated with the event, such as LTE, WCDMA, or GSM.
<code>Events.titles</code>	Event name; the key field for RCA, e.g., Event A3, LTE CA Init, LTE RACH Failure.
<code>Events.description</code>	Detailed raw event description; useful later for selected advanced parsing.
<code>Events.subject</code>	Extra status/context such as Start, End, Success, Failed, or Timeout.
<code>event_timestamp</code>	Parsed event timestamp used for sorting and event-window analysis.
<code>event_delay_from_sample_seconds</code>	Difference between event time and sample time; validates event/sample alignment.

Table 13: Important fields in `events_long.csv`.

Importance and use. The event table connects KPI degradation to actual network procedures. It enables RACH failure RCA, handover analysis, CA effectiveness analysis, WCDMA active-set analysis, call/download event alignment, and event-density anomaly detection.

Cleaning improvement. Keep events in long format. Add derived columns such as `event_family`, `event_severity`, `is_failure_event`, and counts over rolling windows such as RACH failures in the last 30 seconds.

4.3 `event_summary_Events_titles.csv`

This file is the frequency ranking of event titles. It helps identify which event families deserve attention first.

Event title	Count
Event A3	5,758
Event A1	4,980
LTE CA Init	4,365
LTE CA Complete	4,363
UMTS AS Update Attempt	4,187
UMTS AS Update Complete	4,187
LTE RACH Msg1	4,168
Event A6	3,734

Table 14: Top event titles by frequency.

Importance and use. Frequent titles such as Event A3, Event A1, LTE CA Init/Complete, UMTS AS Update, LTE RACH Msg1, Event A6, and RACH Failure should be grouped into mobility, CA, access, WCDMA, and activity families.

Cleaning improvement. Use this table to build the event taxonomy. Rare events should not be deleted automatically because failure events can be rare but important.

5 Planning Validation Output

5.1 drive_nearest_planning_sample.csv

This file attempts to match sampled drive-test points to the nearest planned sector/site. It is useful mainly as a validation check, and it shows that the current planning file is geographically mismatched with the Fiji/Nadi drive route.

Item	Value
Median nearest planned distance	16,056.2 km
Minimum nearest planned distance	16,049.6 km
Rows inside planned radius	0
Unique nearest planned sectors	1

Table 15: Planning-geography validation statistics.

Parameter / Column	Meaning
Location - Latitude / Location - Longitude	Drive-test GPS point.
LTE - Serving - PCI	Observed LTE serving PCI, useful for LTE context but not NR planning validation.
nearest_plan_distance_km	Distance from drive point to nearest planned sector/site.
azimuth_error_deg	Angular difference between planned azimuth and bearing to the drive point.
inside_planned_radius	Whether the drive point falls inside planned cell radius.
plan_site_sector_ID / plan_ siteID / plan_sectorID	Nearest planning record identifiers.
plan_Freq Band / plan_RSI / plan_PCI / plan_Mod4	NR planning attributes from the planning file.

Table 16: Important fields in drive_nearest_planning_sample.csv.

Importance and use. The file proves that the NR planning file should not be used as the main validation source for this route. The nearest planned site is thousands of kilometers away from the drive route, so PCI/RSI comparison would be invalid.

Cleaning improvement. Keep this output as a planning mismatch report. Add an automatic rule: if median nearest-site distance is greater than 20 km, skip planning comparison and warn that the planning file does not match the drive-test geography.

6 Technology Performance Summaries

6.1 lte_summary_by_pci.csv

This file groups LTE samples by serving PCI. It summarizes how each observed PCI behaved in terms of coverage, quality, and throughput.

Item	Value
Unique LTE serving PCIs	190
Total summarized LTE sam- ples	37,588
Highest-exposure PCI	469
Best mean DL throughput PCI (rows ≥ 100)	155

Table 17: LTE PCI summary highlights.

Parameter / Column	Meaning
LTE - Serving - PCI	Observed LTE serving Physical Cell ID.
rows	Number of samples served by that PCI.
first_seen / last_seen	Time window in which the PCI appeared.
mean_rsrp / p10_rsrp	Average and weak-edge LTE coverage strength.
mean_sinr / p10_sinr	Average and weak-edge LTE quality/interference.
mean_dl_tp	Average downlink throughput while served by the PCI.

Table 18: Important fields in lte_summary_by_pci.csv.

Importance and use. This is a core optimization table. It can rank high-impact poor PCIs, suspicious low-throughput PCIs, and benchmark high-throughput PCIs.

Cleaning improvement. PCI alone may mix reused cells. Improve this table later by grouping by PCI + EARFCN + band + route/location cluster. Add percentiles, poor coverage ratio, poor SINR ratio, low throughput ratio, and event counts per PCI.

6.2 wcdma_summary_by_psc.csv

This file groups WCDMA samples by serving PSC. PSC is the WCDMA Primary Scrambling Code and is useful for identifying serving WCDMA behavior along the route.

Item	Value
Unique WCDMA serving PSCs	168
Total summarized WCDMA samples	25,448
Highest-exposure PSC	305
Strongest mean RSCP PSC (rows ≥ 100)	213

Table 19: WCDMA PSC summary highlights.

Parameter / Column	Meaning
WCDMA - ServingCell - PSC	Observed serving WCDMA Primary Scrambling Code.
rows	Number of samples served by that PSC.
mean_rscp / p10_rscp	Average and weak-edge WCDMA coverage strength.
mean_ecno / p10_ecno	Average and weak-edge WCDMA quality/interference.

Table 20: Important fields in `wcdma_summary_by_psc.csv`.

Importance and use. This table is central for the MO call/CSFB branch because MO sessions are WCDMA-heavy. It helps rank weak PSCs, quality-limited PSCs, and healthy benchmark PSCs.

Cleaning improvement. Improve later by grouping by PSC + UARFCN + route/location cluster. Also create `wcdma_cells_long.csv` for active-set and neighbor-cell analysis.

7 Session-Level Output

7.1 session_summary.csv

This table compresses the row-level data into one row per session or unique ID. It is the best control table for splitting the project into LTE download and MO-call branches.

Item	Value
Fiji-Nadi-MO-D	32,048 samples
Fiji-Nadi-LTE-DL-D	31,570 samples
Nawaka-LTE-DL	789 samples
Nawaka-MO	542 samples

Table 21: Rows by route/test label.

Parameter / Column	Meaning
id	Unique session/log/activity identifier.
rows	Number of row-level samples in the session.
time_start / time_end / duration_seconds	Session time coverage and duration.
valid_location_rows	Number of samples with valid GPS.
lte_rows / wcdma_rows / nr5g_rows	Technology availability inside the session.
event_rows	Samples with one or more events.
route_label / location_label / test_type	Parsed metadata from the ID string.
mean_lte_rsrp / p10_lte_rsrp	Session-level LTE coverage summary.
mean_lte_sinr / p10_lte_sinr	Session-level LTE quality summary.
mean_lte_dl_throughput	Session-level LTE throughput summary.
mean_wcdma_rscp / mean_wcdma_ecno	Session-level WCDMA coverage and quality.
lte_pct / wcdma_pct / nr5g_pct / event_pct	Technology/event share inside the session.
min_lat / max_lat / min_lon / max_lon	Approximate session bounding box.

Table 22: Important fields in session_summary.csv.

Importance and use. This table is ideal for session-level anomaly detection, quality scoring, route comparison, and high-level RCA. It proves that LTE-DL sessions and MO sessions should be modeled separately.

Cleaning improvement. Add session-level event counts such as RACH failures, handover

attempts, CA events, WCDMA updates, and script failures. Also add flags for short sessions and sessions with poor radio quality.

8 Cleaned Row-Level Dataset

8.1 drive_clean_base.parquet

The Parquet file is the cleaned row-level dataset. Parquet is a compressed columnar analytics format, which is better than Excel for wide ML/EDA datasets because it preserves data types, loads faster, and stores large tables efficiently.

Why Parquet instead of Excel?

Excel is useful for small reports and manual review, but Parquet is better for the cleaned analytical dataset. It is compact, type-safe, fast to load, and suitable for ML pipelines. Excel may become slow, lose data types, or mishandle long identifiers and timezone-aware timestamps.

This file should be treated as the clean base source for specialized tables rather than the final ML table itself. Recommended derived tables include LTE samples, WCDMA samples, activities, events, locations, sessions, LTE neighbors, and WCDMA cells.

9 How These Outputs Improve the Cleaned Data

Improvement	Reason
Use canonical KPI names	Rename important vendor columns to stable names such as <code>lte_rsrp</code> , <code>lte_sinr</code> , <code>lte_dl_throughput</code> , <code>wcdma_rscp</code> , and <code>wcdma_ecno</code> .
Split by test type	LTE-DL and MO/CSFB sessions represent different problems and should not be mixed in one target model.
Keep sparse event/activity KPIs	Sparse does not always mean useless; many event and activity fields are sparse by design.
Use long event format	<code>events_long.csv</code> is better than packed event strings for RCA and ML features.
Create LTE neighbor long table	Wide neighbor columns should be reshaped into one row per neighbor per sample.
Create WCDMA cells long table	WCDMA active-set/neighbor fields should be reshaped into one row per cell per sample.
Skip invalid planning comparison	The current planning file is geographically mismatched, so it should not drive model conclusions.

Table 23: Recommended improvements to the cleaned data pipeline.

10 Recommended Removed Columns to Bring Back Later

The following columns were removed mostly because they are sparse. They should not all be restored blindly into the main table, but selected ones should be protected in an enriched version of the dataset.

Priority	Column or Group	Reason to restore
High	<code>Throughput.HTTP_DL</code>	Application-layer HTTP downlink throughput. Restore for radio-vs-application throughput anomaly detection.
High	<code>Throughput.Physical_DL</code>	Physical/application-related DL throughput. Useful with <code>HTTP_DL</code> for throughput RCA.
High	<code>Extra_KPIs.UMTS Handover Interruption Time</code>	Sparse but very meaningful for WCDMA mobility and CSFB/call RCA.

Priority	Column or Group	Reason to restore
High	Extra_KPIs.LTE Handover Interruption Time	Important for LTE handover interruption and mobility optimization.
High	Extra_KPIs.ERRC Connection Setup Time	LTE access/setup timing; useful for accessibility and call/download setup RCA.
High	Extra_KPIs.RRC Connection Setup Time	General RRC setup timing; useful for access delay analysis.
High	Extra_KPIs.MO Call Setup Time	Useful target/feature for call setup delay and failed call RCA.
High	Extra_KPIs.MO Call Setup Time Until Connect	More direct call connection timing metric.
High	Extra_KPIs.MO CSFB Call Setup Time	Very important for CSFB voice-call analysis.
High	Extra_KPIs.Attach Delay	Useful for accessibility/session setup analysis.
Medium	Activity.Error	Explains failed activities; should be used in activity-level RCA.
Medium	Activity.TTFB	Time to first byte; useful for HTTP/application-layer RCA.
Medium	Activity.DNS Resolution Time	Application setup delay; useful for separating radio issues from app/server issues.
Medium	Activity.Remote Address / Activity.Remote IPs	Debugging context for HTTP server endpoint issues; not recommended as raw ML features.
Medium	Activity.Size	Activity payload size; important for throughput normalization.
Medium	Activity.ARFCN / Activity.Band / Activity.LCI	Activity-level radio context that helps connect activities to frequency/band/cell conditions.
Special table	LTE NeighborCells(4+) groups	Do not restore as many wide columns; convert to lte_neighbors_long.csv.
Special table	WCDMA WCDMACellsTable(2+) groups	Do not restore as many wide columns; convert to wcdma_cells_long.csv.
Optional	LTE SecondaryCell 2 groups	Restore only if the project includes 3-carrier CA analysis; otherwise keep out of the base table.

Table 24: Recommended sparse columns/groups to restore or reshape later.

Recommended output structure after restoring selected information:

- `drive_clean_base.parquet`: current clean row-level source.

- `drive_clean_enriched.parquet`: clean base plus selected sparse KPIs restored.
- `lte_neighbors_long.csv`: LTE neighbor cells reshaped from wide to long format.
- `wcdma_cells_long.csv`: WCDMA active-set/neighbor cells reshaped from wide to long format.
- `events_long.csv`: event timeline used as the main RCA/event feature table.

11 Final Recommendation

The project should proceed as an LTE/WCDMA QoE anomaly detection and RCA project. The cleanest roadmap is:

1. Use `session_summary.csv` to split sessions into LTE-DL and MO/CSFB branches.
2. Use `drive_clean_base.parquet` as the row-level source.
3. Use `events_long.csv` for event-window RCA and event-derived ML features.
4. Use `lte_summary_by_pci.csv` and `wcdma_summary_by_psc.csv` to prioritize bad PCIs/PSCs.
5. Restore only the selected sparse KPIs needed for application-throughput, handover, call setup, and CSFB analysis.
6. Exclude the current NR planning comparison from the main story until the correct planning file and usable NR measurements are available.